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MILITARY SPECIFICATION

TRUCK, SNOW REMOVAL EQUIPMENT A/S32U-35 (HEAVY DUTY)

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specifications covers a diesel-engine driven, 4-wheel drive, heavy duty snow removal equipment truck having a nominal rated chassis capacity of 46,000 pounds GVW.

1.2 Classification. Trucks shall be of the following type.

A/S32U-35 Equipped with dump body and associated mechanism

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

F-F-351	Filter and Filter Elements, Fluid, Pressure, Lubricating Oil, Bypass and Full Flow
W-C-596	Connector, Plug, Electrical; Connector, Receptacle, Electrical
QQ-C-320	Chromium Plating (Electrodeposited)
VV-F-800	Fuel Oil, Diesel
ZZ-I-550	Inner Tube, Pneumatic Tire

Military

MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-F-3541	Fittings, Lubrication
MIL-H-5606	Hydraulic Fluid, Petroleum Base, Aircraft, Missile, and Ordnance
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-B-11188	Battery Storage, Lead-Acid
MIL-C-25050	Colors, Aeronautical Lights and Lighting Equipment, General Requirements for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODSAC, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-S-83120	Snowplow, Truck Mounting, Power Reversible Blade MXU-512/E
MIL-S-83121	Snowplow, Truck Mounted, Underbody MXK-513/S32U
MIL-S-83122	Snowplow, Truck Mounting, Rollover MXU-514/E

STANDARDS

Military

MIL-STD-12	Abbreviations for use on Drawings and in Technical-Type Publications
MIL-STD-17	Mechanical Symbols (Other Than Aeronautical, Aerospace, and Spacecraft use)
DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-209	Slinging Eyes and Attachments for Lifting and Tying Down Heavy Military Equipment
MIL-STD-281	Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies, Preservation and Packaging of
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MS24333	Coupling Assembly, Hydraulic, Self-Sealing, Quick Disconnect, Flared Fitting to Internal Thread Boss
MS24334	Coupling Assembly, Hydraulic, Self-Sealing, Flareless Fitting to Internal Thread Boss
MS24335	Flange, Bulkhead Mounting, Hydraulic, Self-Sealing Coupling
MS51118	Pintle Assembly, Towing - 40,000 Pound Capacity, Manual Release
MS51317	Light, Warning, Vehicular, Rotating, D.C.

DRAWING

Frink Sno-Plows, Incorporated

3433-1	Hitch, Snowplow, Air Force 4 x 4

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue in effect on date of invitation for bids or request for proposal shall apply.

Tire and Rim Association, Incorporated

Military Supplement to the Yearbook of the Tire and Rim Association, Inc.

(Copies of the TRA Yearbook may be obtained from the Tire and Rim Association,

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Inc., 3200 W Market Street, Akron, OH 44313.)

U.S. Department of Transportation (DOT)

Federal Motor Carrier Safety Regulations

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

Truck Body and Equipment Association, Incorporated

Hoist Classification Chart

(Application for copies should be addressed to Truck Body and Equipment Association, Inc., 401-402 Washington Board of Trade Building, 1616 K Street, N.W. Washington, DC 20006.)

State of California

Vehicle Code of California

(Application for copies should be addressed to the Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95809.)

Society of Automotive Engineers, Incorporated

SAE Standards and Recommended Practices

J551	Measurement of Vehicle Radio Interference (20 to 1000 Mhz)
J588	Turn Signal Lights
J589	Turn Signal Switch
J833	USA Human Physical Dimensions
J910	Hazard Warning Signal Switch

(Application for copies should be addressed to the Society of Automotive Engineers, Two Pennsylvania Plaza, New York, NY 10001.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The truck shall consist of at least the following components:

Item or System Name	See Requirement
Basic vehicle	3.8
Front hitch	3.9
Special lighting	3.10
Electrical wiring	3.11
Snowplow controls	3.12
Snowplow hydraulic system	3.13
Dump body	3.14

3.3 Selection of specifications and standards. Specifications and standards

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for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Recycled and recovered raw materials should be used to the maximum extent possible in lieu of virgin raw materials as long as these materials do not jeopardize the intended use and fully comply with all contract requirements. Materials used shall be free from defects which would affect the performance or maintainability of individual components of the overall assembly.

3.4.1 Thermal insulation material. All insulating material shall be nonflammable, maintain its shape and not settle, sag, pack, absorb moisture, or disintegrate under vibrations, temperatures, and operation encountered in use.

3.5 Design. The basic design of the truck shall be that of a commercial design, four-wheel vehicle having four-wheel drive and two-wheel steer modified as necessary to meet the requirements of this specification. The truck shall be diesel engine driven and shall be equipped with a front hitch as specified herein. The design shall provide for safe vehicle operation at high speeds and for safe, precise day and night time maneuvering in and around airport fixtures and aircraft during conditions of restricted visibility such as during heavy snowfall.

3.5.1 Design for attachments. The truck shall be so designed that the following snowplows will mount on the front hitch:

a. MXU-512/E power reversible blade conforming to MIL-S-83120

b. MXU-514/E rollover conforming to MIL-S-83122

3.5.2 Underbody snowplow. When specified (see 6.2), a MXK-513/S32U Underbody Snowplow conforming to MIL-S-83121 or Root Spring commercial Model F-44 or Wausau commercial Model TG-12 shall be provided and mounted on the trucks along with necessary controls (see 3.12) and associated mechanism.

3.5.3 Reliability. The truck shall have a minimum MTBF (mean-time-between-failures) of 141 hours with a minimum reliability of 0.84 for a 25-hour mission at a confidence factor of 0.90.

3.5.4 Maintainability. The trucks shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with reliability and performance specified herein.

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure.

c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and rapid access unless performance or reliability will be appreciably degraded by the accessible location. In performing maintenance, if engineering reasoning or data

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determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility.

d. Permit maintenance with general-purpose tools and equipment normally available commercially. Use of special-purpose tools and equipment shall be subject to approval by the procuring activity.

e. Minimum number of tools required for maintenance by design practices such as reducing the variety of bolt head sizes to the practicable minimum.

3.5.5 Foolproofness. Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided where practical. That mounting shall allow the item to be installed only in its proper operating position. If an unsymmetrical mounting means it is not practical, the item shall be so mounted that its proper operating position can be readily and visibly determined by service maintenance personnel.

3.5.6 Safety features.

3.5.6.1 Guards. Moving parts that constitute a hazard to operating and maintenance personnel shall be fully enclosed or properly guarded. Exposed parts that are subject to high operating temperatures or that are electrically energized shall be insulated, fully enclosed, or adequately guarded.

3.5.6.2 Nonskid material. All runningboards, steps, and walkways external from the cab shall be of a skidproof design that will permit snow to fall through rather than accumulate.

3.5.6.3 Starting safety device. The trucks shall be provided with a safety device to prohibit engine starting unless the transmission is in a neutral position.

3.5.6.4 Captive fill caps. Attached safety chains shall be provided on all liquid fill caps to prevent loss or misplacement of same.

3.5.6.5 Cab access. Cab accessibility shall be such that the path which personnel will follow on entering or leaving the cab from either side is direct, unobstructed, and will not intersect the plane of the lowered dump body head sheet. Sufficient handholds and steps shall be provided to enable male personnel, with physical dimensions in the range of small to large as defined in SAE Standard J833, wearing heavy winter clothing to easily and safely enter the cab.

3.5.6.6 Operator's vision. The trucks shall be so designed that when equipped with hitch, but without front mounted attachments and with the exception of a windshield centerpost, the operator's view of the road and adjacent areas throughout the entire windshield area shall not be restricted. A small to large operator, as defined in SAE Standard J833, when sitting in an erect normal sitting position shall have full forward view of the ground.

3.5.6.7 Safety belts. Two safety belts shall be so installed that the yield strength of the connection is equal to or greater than the rated ultimate strength of the belt.

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3.5.6.8 Grommets and clamps. Grommets or clamps shall be provided to prevent chafing where wire, tubing, and hose pass through structural members.

3.5.7 Lifting and tiedown rings. The truck shall be provided with lifting and tiedown rings in accordance with MIL-STD-209 located at accessible points that will permit cable attachment and approximate level hoisting without damage to the truck or its components. The rings shall be so located on the truck that transportation personnel can rig safe lifting slings from common cable and spreader bar components for lifting by a single-hook, overhead hoisting facility.

3.5.7.1 Component lifting provisions. Lifting eyes or other hoisting devices shall be provided on or for attachment to major components so that they can be readily removed or replaced on the truck by means of overhead hoisting facilities.

3.5.8 Fittings. Disconnect plugs and protected receptacles, multiple line connectors in the electrical system, and readily detachable and attachable type fittings in hydraulic and pneumatic systems, as applicable, shall be utilized wherever practicable to permit rapid component removal and replacement. All disconnect points shall be readily discovered.

3.5.9 Lubrication. The trucks shall be designed and constructed to operate satisfactorily when lubricated with the standard military lubricants specified in MIL-M-8090.

3.5.9.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541 and shall be located in accessible, protected positions. Extended fittings shall be provided to lubricate parts or assemblies that are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibilty.

3.5.9.2 Lubrication chart. A lubrication chart shall be provided directing attention to all lubrication fittings and shall specify the range and grade of lubricant required for critical temperatures. The chart shall be permanently attached to the unit in an accessible and convenient location.

3.5.9.3 Drain plugs. All plugs installed for the purpose of draining lubricants shall be of the permanent magnet type.

3.6 Construction. The truck shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, and service.

3.6.1 Standard vehicles and accessories. Except as specified herein, vehicles, components, assemblies, and accessories to be delivered under the contract shall be standard or optional items, which meet or exceed the requirements of this specification.

3.7 Performance.

3.7.1 Environmental conditions. The truck shall meet the following environmental conditions without detrimental effect to subsequent operation:

a. Starting and operation in ambient temperature ranging from:

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(1) -20° to +125°F without the use of external heat or installed power plant heaters (fluid starting aids may be used)

(2) -40° to +125°F using the installed power plant heaters (see 3.8.2.8)

b. Storage in ambient temperatures from -65° to +160°F.

3.7.2 Mobility. The trucks, loaded and equipped as specified in 3.8.1, shall attain the following minimums on dry, clean, concrete or asphalt surfaces. These minimums shall be obtained with a torque converter efficiency of not less than 70 percent.

a. Vehicle speed, drawbar pull, and gradeability:

DIRECTION	SPEED (MPH)	DRAWBAR PULL (POUNDS)	GRADE (%)
Reverse	8	Not required	0
	0 (static at wheel slip)	25,000	0
	5	12,000	0
	10	8,200	0
	15	5,500	0
	20	4,000	0
	25	3,200	0
	30	2,800	0
	35	Not required	5.0
	50	Not required	2/5

b. Turning radius to the right of left of 35 feet (see 6.3.3).

3.7.3 Operation. The truck shall be capable of operating under the conditions specified herein for at least 4,000 miles without failure.

3.8 Basic vehicle.

3.8.1 Chassis. The chassis shall be of a four wheel, all wheel drive, two wheel steer configuration with single tires front and rear. The gross operating weight shall include the weight of the chassis and cab with all accessories and equipment, hitch, full complement of fuel, lubricants and coolants, two operators, and the dump body with ballast evenly distributed over the load area to provide an operating weight of 46,000 pounds (see 3.14). The load on either axle shall be not less than 45 percent nor more than 55 percent of the gross operating weight. Chassis component ratings shall not be arbitrarily raised to meet these operating conditions. The drive train components including but not limited to the torque converter, transmission(s), propeller shafts, axle shafts, and differentials shall be matched to the engine, shall be compatible, and shall transmit the maximum engine torque as multiplied by the torque converter through all reductions. The drive train components shall be of a commercial design and capacity that has been produced and supplied as a part of a commercial vehicle. Unless otherwise specified, the spare wheel assembly and wheel changing tools shall not be provided.

3.8.2 Vehicle engine. The engine shall be a liquid-cooled commercial standard diesel capable of developing and maintaining sufficient horsepower for efficient

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operation without exceeding 85 percent of the engine manufacturer's certified maximum automotive duty net rating of the engine at sea level and at 60°F. This operation shall be obtained with the engine using fuel conforming to VV-F-800. Installation of the front mounted hitch and cab shall assure adequate consideration for engine maintainability as described in 3.5.4c.

3.8.2.1 Governor. An engine governor shall be furnished, set and sealed to limit the engine to the maximum recommended operating speed.

3.8.2.2 Generator and starting system. Trucks shall be furnished with 24-V starting systems, with 12-V lighting system, and voltage regulators. The rated amperage capacity of the generator or alternator provided shall be not less than the sum of the average current draw of all constant load components plus 35 percent of the average current draw of all intermittent load components. At normal engine idle speed, not less than 20 amp dc output shall be provided.

3.8.2.3 Air cleaner. An air cleaner of the dry, pleated, phenolic resin-treated paper type or an oil bath type with an oil capacity of not less than 1 quart shall be furnished.

3.8.2.4 Oil filter. Oil filter shall be full-flow type or bypass type with filter element conforming to F-F-351.

3.8.2.5 Heavy-duty cooling system. A heavy-duty cooling system shall be furnished which shall maintain coolant at a temperature below the boiling point with the truck loaded to rated GVW and operated in an ambient air temperature of not less than 125°F at sea level.

3.8.2.5.1 Radiator. The radiator shall be equipped with a thermostatically controlled shutter. Means shall be provided to permit manual operation. The front shall remain in the open position upon failure of the thermostatic control.

3.8.2.6 Fuel tank(s). Fuel tank(s) shall be the manufacturer's standard for the type vehicle specified. The fuel tank(s) shall be of adequate capacity to provide for 8 hours or more of full load operation.

3.8.2.7 Exhaust pipe. When a vertical type engine exhaust stack is used, it shall be positioned to the rear of the cab and provided with a positive closing flip-top cover. A 1/4-inch hole shall be provided at the base of the exhaust stack to drain exhaust condensate.

3.8.2.8 Power plant heater kit. Unless otherwise specified (see 6.2), a power plant heater kit shall be installed on the truck prior to delivery. The kit shall provide standby-type winterization and shall include properly selected 110-V heaters and thermostats. Heaters shall be of sufficient capacity to place the truck in an operating condition at -40°F within 60 minutes.

3.8.2.8.1 Heater connections. A recessed, male, three-prong, 110-V receptacle conforming to style D or X of W-C-596 for the power plant heaters shall be mounted and placarded on the outside of the cab. Two prongs of the receptacle shall be connected to the winterization kit heaters and the third prong shall be grounded to the chassis. The receptacle shall be in a protected location and shall have a spring-loaded dust cap included as part of its housing. A three-wire connecting cable, 25 feet long commercial SO, of adequate line capacity to supply power for all heater units simultaneously shall be furnished.

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Connecting cable shall include a matching female connector at the vehicle end and a standard, weatherproof, three-pronged (two power plus one ground) male connector at the other end. A carrier for the connecting cable shall be mounted within the cab or engine compartment and shall provide positive cable retention during vehicle operation.

3.8.3 Main transmission. The truck shall be provided with a semiautomatic transmission in which limited continuous variable torque ratio changes occur automatically and in which further required ratio changes must be controlled by the operator with only instantaneous interruption of power flow with each selection of ranges. The transmission shall provide for driver selection of individual gears or selection of ranges of gears required for efficient operation throughout the speeds specified herein. The shift control shall incorporate a positive indicator for each gear or range selected. Thr transmission shall be so protected that the operator cannot inadvertently shift the truck into reverse. Hand- or foot-actuated clutch members shall not be used. The transmission shall be so designed that it can be shifted while the maximum torque of the engine, as multiplied by the torque converter, is being delivered. The time limit for full engagement shifting of forward ratios under full throttle shall not exceed 1.5 seconds. To facilitate rocking the vehicle back and forth, the time limit for full engagement shifting between forward and reverse under full throttle shall not exceed 2 seconds. Means shall be provided for towing the vehicle, with the engine inoperative, without damage to the transmission. When the torque converter is subjected to 45 seconds of wide-open-throttle stalling, the oil temperature shall not exceed 265°F from an initial temperature of not less than 130°F and an ambient temperature of not less than 30°F.

3.8.3.1 Auxiliary transmission. An auxiliary two-speed transmission, if necessary to meet performance requirements, shall be located after the main transmission in the power train of the vehicle. The shift control shall incorporate a positive indicator for each gear and include a neutral position. Foot-actuated clutch members shall not be used.

3.8.4 Axles. Axle capacity shall be at least equal to the load imposed on each axle, measured at the ground, when the vehicle is loaded to rated GVW and with or without the respective front mounted attachments installed and in the raised position. The ballasted weight shall be as described in 3.8.1 and the design weight of the front mounted attachment will weigh 4,000 pounds, 52 inches forward of the hitch mounting pin. Gear ratio shall provide performance as specified in 3.7.2. The driving-steering axle shall incorporate sealed universal joint housings and steering universal joints to minimize any change in velocity throughout their range of angular travel. Axle capacity shall be based on the axle manufacturer's certified capacity rating.

3.8.4.1 Suspension. Vehicles shall be equipped with suspension system with components having a rated capacity at least equal to the load imposed on each member, measured at the ground, and matching the axle capacity as determined in 3.8.4. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. Hydraulic, double-acting shock absorbers shall be provided on the front axle.

3.8.5 Differentials. Differential devices shall be provided between wheels and axles. The devices shall provide for differences in speed between wheels on each axle and differences in speed between axles while insuring that each wheel in the rear axle and the transfer case will receive and transmit a per-

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centage of the drawbar pull capability in proportion to the slip torque capability of that individual wheel. The differential devices shall not disengage while turning or backing the vehicle.

3.8.6 Wheels, tires, and tubes. The trucks shall be equipped with single front and single rear wheel assemblies of the same size. Wheels and highway type pneumatic tires (tube or tubeless) having an open diamond type tread which will not pick up stones while operating off paved surfaces and deposit them on paved surfaces shall be provided, and shall meet the mobility requirements of 3.7.2. The front and rear wheels shall be interchangeable. The tires shall be rated in accordance with the Tire and Rim Association Yearbook. The tire ratings shall be at least equal to the imposed load measured at the ground when a front mounted snow removal attachment is not attached to the ballasted truck (see 3.8.1). With the truck in this configuration and a load on the hitch equal to 4,000 pounds which has a center of gravity of 52 inches forward of the hitch mounting pins, the load imposed on each tire during vehicle operation with the front attachment in the raised position shall not exceed 5 percent of its rated capacity as indicated in the Tire and Rim Association Yearbook. When specified (see 6.2), studded tires shall be furnished.

3.8.6.1 Wheel assemblies. All wheels assemblies shall be balanced in accordance with standard practice normally used by truck manufacturers for similar vehicles furnished for use under similar operating conditions.

3.8.6.2 Traction devices. The trucks shall be so constructed that there is sufficient clearance to permit satisfactory installation and use of tire chains on all wheels.

3.8.6.3 Inner tubes. When tube type tires are furnished, inner tubes shall be of heavy duty type conforming to ZZ-I-550, and shall be of proper size for tires furnished. Tire flaps shall be provided for tube type tires in accordance with the Tire and Rim Association recommendations.

3.8.7 Cab. The cab configuration and location of the hitch with respect to the cab shall be such as to allow visibility as specified in 3.5.6.6 and operation of the MXU-514/E attachment. Weather stripping shall be provided on the cab doors, windows, and vents to prevent the entry of fine blowing snow or water.

3.8.7.1 Cab heater. A standard commercial heavy-duty hot water personnel heater shall be incorporated and shall provide a cab temperature of at least 45°F in a -45°F environment within 60 minutes after the vehicle engine is started. Defroster outlets with independent blowers shall be provided and installed to provide a flow of heater air to each section or side of the windshield.

3.8.7.2 Seats. The driver's and assistant driver's seats shall be a commercial bucket type or full width or combination thereof adjustable longitudinally and vertically and equipped with an adjustable tilt backrest.

3.8.8 Miscellaneous chassis provisions.

3.8.8.1 Pintle assembly. A pintle assembly conforming to MS51118 shall be mounted at the rear of the truck. Mounting shall include reinforcement to transfer pintle loads directly to chassis side rails. The rear edge of pintle shall be installed not more than 4 inches forward of the rearmost part of the

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vehicle.

3.8.8.2 Brakes. Brakes shall conform to Federal Motor Carrier Safety Regulations 393.40 through 393.43, 393.45 through 393.52, and as specified herein, except the stopping distance shall be 30 feet from an initial speed of 20 mph.

3.8.8.2.1 Air or air hydraulic type. The truck shall be equipped with four-wheel, full-air-application or air hydraulic type brakes. The braking system, complete with all necessary components, shall include:

a. Air compressor, unloader-head-type, engine-driven and engine-lubricated, air or water cooled, and having a capacity of not less than 7-1/4 cfm

b. Air storage reservoir(s) with not less than 2,200 cubic inches total capacity, each tank equipped with drain, and safety and check valves between compressor and last reservoir tank

c. Foot control, suspended or treadle type

d. Limiting valve for front brakes, combined with quick-release valve (full air system only)

e. Quick-release valve, for rear brakes (full air system only)

f. Air pressure gauge, visible to the driver

g. Low-air-pressure warning, visible and audible

h. Alcohol aspirator or moisture ejector.

3.8.8.3 Air brake emergency system. When specified for full-air application brakes (see 6.2), an emergency air brake system capable of being applied by the driver shall be provided. The air brake emergency system shall automatically apply the brakes at an air pressure of not less than 20 psi nor greater than 45 psi, or a split modulated emergency brake system shall be provided. When compressed air is not available to release the brakes, a manual or mechanical release shall be provided. All components and the complete system shall conform to the California Vehicle Code, Section 26508.

3.8.9 Battery and battery compartment. When specified by the procuring activity (see 6.2), dry charged batteries and electrolyte shall be furnished separately. The battery compartment shall be vented to prevent gaseous accumulations. The batteries furnished shall be of a type specified in MIL-B-11188. Batteries shall be contained in a properly designed metal box or frame in a protected, easily accessible location.

3.8.10 Lighting. All lights, reflectors, and wiring shall be as specified herein and shall be in accordance with Federal Motor Carrier Safety Regulations 393.12, 393.13, 393.19 through 393.26(c) and 393.27 through 393.33. Lights and reflectors shall not be mounted on rub rails or bumpers.

3.8.10.1 Turn signals. Turn signal units SAE standard J588, class A, with SAE standard J589 class A operating units mounted on the steering column shall be furnished. The turn signal operating units shall be of the self-cancelling type with visible and audible flash indicators. A vehicle hazard warning signal in accordance with SAE J910 shall be incorporated in the turn signal

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operating unit. The vehicle shall be provided with double-faced front signal units and single-faced rear signal units.

3.8.11 Steering. Power steering shall be furnished on all trucks. Failure of the power steering shall not prevent manual steering.

3.8.12 Windshield wipers and washers. The trucks shall be equipped with dual windshield wipers and windshield washers. Windshield wipers shall be multi-speed type and operated by either vacuum with booster, air, or electric motor(s). The windshield wipers shall be mounted for operation from above the top portion of the windshield. Either a dual windshield washer system or a single system with a nozzle for each side of the windshield will be acceptable.

3.8.13 Vehicle controls and operating mechanism. All vehicle controls and operating mechanisms shall be located for left hand drive. Controls shall be complete and conveniently operable by the driver. Lever controls shall be designed and located to permit easy entrance and exit of operator to and from the driver's compartment. Instruments and controls shall be identified as to their function and installed in a manner to facilitate removal and servicing. Instruments shall be panel mounted.

3.8.14 Accessories and equipment. Chassis equipment shall be complete with all accessories furnished as standard equipment. At least the following equipment shall be furnished for all vehicles:

a. Outside rearview mirrors, mounted on each side of cab, each with at least 85 square inches of reflective area; each with at least three supporting arms

b. Ignition switch (not key-actuated)

c. Ammeter or charging indicator

d. Fuel gauge

e. Oil pressure gauge or indicator

f. Engine temperature gauge or indicator

g. Speedometer with recording odometer

h. Ash receptacle

i. Cigar lighter

j. Dual sunvisors

k. Front door mounted arm rest on driver's side

l. Driver's compartment ventilator other than window

m. Standard electric, or air, horn for this size equipment.

3.9 Front hitch. The truck shall be equipped with a snowplow hitch conforming to Drawing 3433-1. The clearance between the bottom of the hitch and the ground, with the vehicle ballasted to 46,000 pounds and equipped with any com-

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bination of attachments listed under 3.5.1, shall be not less than 11 inches nor more than 13 inches. All front-mounted snowplows shall connect to this hitch.

3.9.1 Truck mounting brackets. The truck mounting brackets shall be essentially as shown on Drawing 3433-1 with design changes as required to mount the hitch on the truck and to provide clearance for operation of the MXU-514/E attachment.

3.10 Special lighting.

3.10.1 Headlights. The headlights shall be so positioned that the front-mounted attachments cannot intersect the light beam.

3.10.2 Spotlights. A spotlight shall be provided on each side of the forward part of the cab. The spotlights shall be operable from inside the cab. A PAR-46 lamp, trade No. 4577, shall be installed within each spotlight housing.

3.10.3 Warning light. A 24-V warning light dimensionally conforming to MS 51317 shall be installed on the top of the truck. Unless otherwise specified (see 6.2), the color shall be aviation yellow as specified in MIL-C-25050. A switch shall be provided on the instrument panel to control the light. The light assembly furnished shall be mounted for easy removal and installation. The light shall be visible 360° around the truck and shall meet the following characteristics:

a. Unless otherwise specified (see 6.2), the light shall be of the rotating type with a flash rate of 60 to 100 flashes per minute when observed from a fixed position.

b. A minimum light intensity in all vertical planes, expressed in terms of the effective candle-power, equal to or exceeding the following values:

LIGHT INTENSITIES (WITH SPECIFIED COLOR FILTERS)

Angle Above or Below Horizontal in Degrees	Minimum Effective Candle-Power with Specified Color Filter
5 above to 5 below	100
5 to 10 above	90
10 to 20 above	70
20 to 30 above	50

3.11 Electric wiring. Electric wiring shall be in accordance with SAE standard J555.

3.11.1 Circuit marking. The circuits shall be color coded or numbered.

3.11.2 Protective devices. Overload circuit breakers with manual resets shall be used in lieu of replaceable, fuse type devices. These and any other protective devices shall be of an approved type and provided as required to properly protect the systems. Protective devices shall be connected and identified in an organized manner to facilitate determination of malfunction.

3.12 Snowplow controls. The trucks shall be equipped with all controls as required for operating the snow removal attachments specified in 3.5.1 and

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the underbody snowplow specified in 3.5.2 (when so equipped). Hydraulic, air over hydraulic, or electric hydraulic controls shall be centrally located for operation by the driver with his seat belt fastened. The controls shall be arranged to avoid confusion to the driver and the operator and shall be identified as to function and use by nameplates. Control switches shall be spaced to permit operation while wearing heavy mittens or gloves and shall be located from left to right as follows:

Number Required	Function
1	Raise and lower front-mounted attachments
1	Lock - reversible blade
1	Rotate - rollover snowplow Rotate - reversible blade
1	Raise and lower - snowplow deflector
1	Underbody snowplow - horizontal rotation 1/
1	Underbody snowplow - moldboard tilt, L.H. 1/
1	Underbody snowplow - moldboard tilt, R.H. 1/
1	Underbody snowplow blade - vertical to horizontal rotation 1/
1	Underbody snowplow - horizontal rotation - lock 1/

1/ When the underbody blade is provided.

3.13 Snowplow hydraulic system. The hydraulic system for operation of the attachments listed in 3.5.1 shall consist of rams, pump, supply tank, hoses, piping, fittings, valves, controls, and other parts essential to practical and efficient operation. The hydraulic system shall be designed for use of, and filled with the proper amount of fluid conforming to MIL-H-5606. A minimum number of fittings, joints, and connections shall be used to prevent excessive backpressure, clogging, and leakage. The pump shall be driven by the vehicle engine. The driving mechanism shall not slip when wet. The pump shall have sufficient capacity to provide satisfactory simultaneous operation of any combination of hydraulic components and attachments. All hydraulic tubing or hose shall be of a size that will permit free flow of the hydraulic fluid at temperatures down to -40°F. Provisions shall be made for completely draining all hydraulic lines.

3.13.1 Hydraulic equipment. Under any operating condition, the duty cycle hydraulic pressures at any point in the system shall not exceed 2,000 psi, nor shall the operating temperature exceed 225°F. All seals, rings, gaskets, et cetera, shall be designed for easy removal and replacement. Piston rams shall covered or protected to provide for operation throughout the climatic environment. Rams and interior cylinder surfaces shall be plated with type II chromium in accordance with QQ-C-320. Metal tubing shall be used in hydraulic lines where flexing is not encountered during operation. Armored or otherwise protected rubber or synthetic hose shall be used where flexing is encountered. Provisions shall be made for bleeding the system at all necessary points. Hydraulic hoses and hose assemblies shall be in accordance with good commercial practice.

3.13.2 Supply tank. The hydraulic fluid tank shall have adequate capacity to hold the maximum volume of fluid for any combination of attachments that will be required in the system during any operating condition plus 10 percent and an additional tank capacity of 10 percent for air space to handle expansion, foaming, and surging of fluid in the tank. The tank shall be located where it can be easily filled directly from a standard gallon fill can. The tank shall

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be provided with a filler opening of not less than 1-1/2 inches in diameter. A vented filler cap and a removable strainer shall be provided. The filler cap shall be designed to prevent fluid from splashing out of the tank during movement. A filter and shutoff valve shall be provided to facilitate cleaning of the fluid and tank. The drain shall be so located that the fluid can be recovered in a container and not drain onto other components. The tank shall be placarded to show requirements for use of fluid conforming to MIL-H-5606 and the fluid level to be maintained . A graduated bayonet or dip stick shall be provided to determine the fluid level and shall be located in a readily accessible location.

3.13.3 Junction panel. A junction panel for front mounting plow attachments shall be mounted on the front right-hand corner of the trucks. The panel shall have self-sealing quick-disconnect couplings conforming to MS24333-8 or MS24334-8. The couplings shall be secured to the plate by a flange conforming to MS24335-8. Nameplates and chain-secured dustcaps shall be provided for each coupling. An instruction plate detailing the hook-up procedure(s) shall be provided near the junction plate.

3.14 Dump body system for A/S32U-35 truck. The A/S32U-35 snow removal truck shall be furnished with a dump body system as follows:

3.14.1 Dump body. The dump body shall have a 7 cubic yard water level capacity to contain the ballast required in loading to an operating weight of 46,000 pounds. Body floor, sides, tailgate, and front head panels shall be constructed of metal having a strength at least equal to that of 8-gage (0.1644 inch) steel. Full length, channel formed, rub rail of minimum width to cover rear tire treads shall be provided. Not less than three triangular or box section side braces per side shall be equally spaced between the body front and full box-type rear corner posts and welded sideplates. The side braces shall be constructed of metal having a strength at least equal to that of 10-gage (0.1345 inch) steel. The front head sheet shall be formed or reinforced for rigidity. The front head and tailgate shall be not less than 8 inches higher than the side. Sides shall have pockets provided at each end for insertion of sideboards. All welding of the body shall be continuous. A cab protector shall not be provided.

3.14.2 Tailgate. The tailgate shall be double-acting, opening fron the top or bottom. The tailgate shall have heavy-duty hardware, heavy-duty support chains, and a heavy-duty tailgate latch operable by a control at the left front corner of the dump body. The tailgate shall be reinforced to prevent deformation under load.

3.14.3 Hydraulic hoist. The hoist shall be the manufacturer's standard with a minimum lifting capacity rating in accordance with the Truck Body and Equipment Association Hoist Classification Chart. The hoist shall lift the body to a minimum dumping angle of 50° from the top of the truck chassis frame. The valve control shall be a commercial standard for the hoist model furnished and shall be located in an accessible location inside the cab.

3.14.4 Understructure. Body longitudinal sills shall be provided to support the hoist load. When the body floor is constructed in two or more pieces, an automatic seam weld having full penetration will be acceptable without reinforcement. When the seam is hand-welded, the underside of the joint shall be reinforced with 3-inch wide, 8-gage (0.1644) butt plates welded between cross-members.

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3.14.5 Body mounting. The dump body shall be mounted to the chassis in accordance with the manufacturer's standard commercial practice. The clearance between the rear of the cab and the dump body front head panel shall be not less than 12 inches. This clearance shall be maintained outside of the chassis frame.

3.15 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.16 Electromagnetic interference. The trucks and all components shall comply with the interference requirements specified in SAE J551.

3.17 Dimensions. Limiting truck dimensions or characteristics are as follows:

a. Maximum overall width - 96 inches

b. Maximum overall height with spotlights and warning lights - 128 inches

c. Tread width - front and rear shall be plus or minus 1 inch

d. Pintle hook height - 30 inches minimum, 48 inches maximum

e. Angle of approach, breakover, and departure (without plow attachments) -20°.

3.18 Finishes and protective coatings. Cleaning, painting, and finishing shall be in accordance with MIL-STD-808. The exterior color shall be yellow, film designation DG. The top of the hood shall be black, film designation UL.

3.19 Markings.

3.19.1 Transportation data plates. Transportation data plates sized and marked as required in MIL-P-514 shall be provided. All other data plates, placards, charts, instruction plates, et cetera, shall conform to MIL-P-514, size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12. Symbols shall be in accordance with MIL-STD-17.

3.19.2 Operational markings. The operational markings shall be applied in accordance with the marking requirements of MIL-STD-808. Stenciling shall be in accordance with MIL-STD-808.

3.19.2.1 Tire pressure. The tire inflation pressure shall be stenciled on the truck as near each wheel as practicable.

3.19.3 Identification of controls. All valves, gages, controls, and indicators other than standard vehicular instrumentation used in the operation of the truck shall be identified by securely attached nameplates of such composition that exposure to oil, dirt, light, et cetera, will not cause them to fade or become eradicated. Tags or decals shall not be used.

3.19.4 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed on or near the control panel. The instructions shall be clear, concise, and adequate to enable operation without damage to the equipment or injury to personnel. Instruction panels shall be of sheet

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aluminum or sheet zinc of not less than 0.050 inch thickness, anodized or etched to produce raised or depressed markings with a black or other dark color background, and with a border of not less than 1/4 inch.

3.19.4.1 The following instruction plates shall be installed in conspicuous locations on the dash panel of the cab:

CAUTION

THIS TRUCK IS EQUIPPED WITH ELECTROMAGNETIC INTERFERENCE SUPPRESSION DEVICES. CARE SHOULD BE EXERCISED DURING MAINTENANCE TO RETAIN AND REINSTALL THESE DEVICES PRIOR TO REUSE OF THE UNIT.

3.20 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.21 Cleaning. The truck shall be thoroughly cleaned and loose weld, spattered or excess solder, metal chips, and other foreign material removed during and after final assembly.

3.22 Workmanship. Defective components or parts and assemblies which have been repaired and modified to overcome deficiencies shall not be furnished.

3.22.1 Steel fabrication. Steel used in the fabrication of equipment shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting, using a tip suitable for the thickness of metal, may be employed instead of shearing or sawing. Re-entrant cuts shall be made in the best possible manner. All bends of a major character shall be made with controlled means to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly.

3.22.2 Bolted and riveted connections. Bolt and rivet holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight. Rivet heads, when not countersunk or flattened, shall be of approved shape and of uniform size for the same diameter of rivet. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member.

3.22.3 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Weld penetration shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Welding shall be in accordance with commercial automotive practice.

3.22.4 Machine work. All parts shall be manufactured to gage through the use of proper jigs and fixtures, and shall permit interchangeability of parts so manufactured.

3.22.5 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or other defects which might render the casting unsound for use.

3.23 Servicing and adjusting. Prior to acceptance of the vehicle by the

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Government inspector, the contractor shall service and adjust each vehicle and its mounted equipment for operational use, including at least the following: focusing of lights, adjustment of engine, adjustment of electrical and brake systems, filling and charging of batteries (except where dry charged batteries are specified), front end alignment (steering), inflation of all tires, complete lubrication of running gear components and chassis including engine and mounted equipment with the grades of lubricants recommended for the ambient temperature at the destination point, servicing of cooling system with a solution of ethylene glycol and water in equal parts by volume, and servicing of windshield washer reservoir with anti-freeze solution.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the truck shall be classified as follows:

a. Preproduction testing . . . . . See 4.4

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions.

4.3.1 Prior to performance testing, all adjustments of brakes, throttle linkage, tire inflations, and ignition timing shall be made in accordance with the manufacturer's recommendations. When practical, specified attachments shall be mounted on the vehicle and tested concurrently in accordance with the applicable specification. The operating weight shall be adjusted to simulate maximum tire and axle loadings encountered under the conditions specified in 3.8.1.

4.4 Preproduction testing (see 6.2).

4.4.1 Test sample. The test sample shall consist of a model representative of the production equipment.

4.4.2 Test report. After completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished to the procuring activity. Particular attention shall be given to recording and maintaining a log of time and manhours required for all preparation, testing, inspection, and repair.

4.4.2.1 Reliability and maintainability information. the following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. Test conditions

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at the time of the events identified shall be recorded

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made or proposed to be made to production items. The summary shall also include an analysis of the predicted effectiveness of such corrections

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions shall be included

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.4.3 Preproduction tests. The preproduction tests shall consist of all tests described under 4.6.

4.4.3.1 Truck equipped with MXK-513/S32U underbody blade. When equipped with the MXK-513/S32U underbody blade, the preproduction tests of MIL-S-83121 shall be conducted in conjunction with the preproduction tests required herein.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual tests . . . . . See 4.5.1

b. Sampling plan and tests . . . . . See 4.5.2.

4.5.1 Individual tests. Each truck shall be subjected to the following tests as described under 4.6:

a. Examination of product . . . . . See 4.6.1

b. Five-mile test . . . . . See 4.6.2.

4.5.2. Sampling plan and tests

4.5.2.1 Periodic sampling. One truck shall be selected at random from every 50 or fraction thereof produced and subjected to the following tests as described under 4.6:

a. Highway speed test . . . . . See 4.6.3

b. Brake tests . . . . . See 4.6.4

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c. Turning test . . . . . See 4.6.5.

4.5.2.1.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of the failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.1.2 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a test failure. But final acceptance of the items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods.

4.6.1 Examination of product. The truck shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, dimensions, and marking, and to determine that the truck is in operable condition.

4.6.2 Five-mile test. The running-in of the truck and all equipment thereof shall be accomplished in accordance with the manufacturer's recommendations; however, the truck shall be driven at least 5 miles over secondary roads. The truck shall exhibit no malfunction, leaks, rough operation, vibration, or other irregular operation as a result of being driven this distance.

4.6.3 Highway speed. The truck ballasted to 46,000 pounds operating weight shall be tested to determine its maximum speed over a distance of not less than 1,000 feet and grades as specified in 3.7.2. Lesser grades may be used providing a drawbar horsepower is developed equivalent to that used in climbing the greater grade. A minimum of three runs shall be averaged. The speed shall be 50 and 35 mph, respectively, or greater, determined by either a speed measuring wheel or as computed by time-distance formulas.

4.6.4 Brake tests. The following tests shall be conducted on dry, clean concrete or asphalt surface.

4.6.4.1 Service brakes. The truck ballasted in accordance with 3.7.2 shall make 10 successive stops from a speed of not less than 20 mph at time intervals not exceeding 1 minute. The brakes shall be allowed to cool for 5 minutes and the above stops repeated. The truck shall come to a complete stop each time within 30 feet after initial movement of the brake pedal. There shall be no evidence of lining breakage, excessive lining or brake-drum wear, brake chatter, grabbing, or squealing. Tests shall be conducted at an ambient temperature of not less than 40°F.

4.6.4.2 Parking brake. The following test shall be conducted subsequent to the test specified in 4.6.4.1.

4.6.4.2.1 The parking brake shall be applied and a towing force sufficient to

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move the truck forward and backward shall then be applied. The wheels shall skid and not roll.

4.6.5 Turning test. The truck shall be operated through one full 360° turn in each direction. The turning radius need not be concentric but must be equal to or less than 35 feet (see 6.3.3).

4.6.6 Drawbar pull. The truck equipped and ballasted as specified in 3.7.2 shall be tested on dry, level, clean concrete or other suitable surface to insure that the drawbar pull conforms to the minimum requirements of 3.7.2. Measurements shall be obtained by use of a chassis dynamometer, a mobile dynamometer, or strain-gage towbar, and fifth-wheel-type speedometer. A minimum of four runs shall be averaged.

4.6.7 Vehicle power train components. The following tests shall be conducted on dry, clean, level concrete or other suitable surface to assure that all drive train components are of sufficiently rugged design. The truck shall be equipped and ballasted as specified in 3.8.1. After completion of these tests, the truck shall be inspected as specified in 4.6.8.2.

4.6.7.1 Vehicle drive train. The truck shall be connected from the rear pintle to an immovable object through a strain-gage drawbar and the maximum drawbar pull under each of the following conditions shall be recorded. A minimum of three runs shall be averaged. The drawbar pull obtained under conditions b through f shall be not less than the percentage of the gross vehicle weight over the driving wheel(s) times the drawbar pull value obtained under condition a.

a. All four wheels driving

b. Front wheels on heavily greased metal plates

c. Rear wheels on heavily greased metal plates

d. Right side wheels on heavily greased metal plates

e. Left side wheels on heavily greased metal plates

f. All four combinations of three wheels on heavily greased metal plates.

NOTE: Rollers may be used in lieu of the heavily greased metal plates.

4.6.7.2 Truck rocking test. The following test shall be conducted to verify the ruggedness of the transmission and the torque reversal capacities of other driveline components. The transmission shall be alternately shifted from lowest forward gear to reverse gear at 10-second intervals for 2 minutes. Vehicle brakes shall not be used during this test. The shifting times shall be as specified in 3.8.3. Immediately following each shift, the engine shall be brought to wide-open throttle for 5 seconds then released for 4 seconds and the gear shifted to the opposite direction of travel. Two additional gear changes, from lowest forward gear to reverse gear and from reverse gear to lowest forward gear, shall be made at wide-open throttle as follows: From standstill, the truck shall travel a distance of at least 20 feet and the gear shifted to the opposite direction with wide-open throttle maintained until the truck changes direction. Shifting difficulties shall not be encountered. The oil temperature shall not exceed 265°F from an initial temperature of not less than 130°F

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and an ambient temperature of not less than 30°F.

4.6.7.3 Stall test. The torque converter shall be stalled for a period of not less than 45 seconds with the engine operating at wide-open throttle. The torque converter oil temperature shall not exceed 265°F from an initial temperature of not less than 130°F in an ambient temperature of not less than 30°F, except a 1°F increase above 265°F shall be allowed for each degree ambient temperature above 30°F. Resumption of forward movement at a speed not exceeding 5 mph shall be sufficient to cool the torque converter oil back to a stabilized operating temperature within 15 minutes.

4.6.8 Endurance tests. The truck shall be subjected to 4,000 miles of operation over a test course to determine capabilities under sustained usage. The test shall consist of four 1,000-mile cycles. Items a through i shall be conducted with the vehicle ballasted as specified in 3.8.1. If driver controlled differentials are provided, they shall be in locked condition during the test specified under item c. Adverse weather conditions shall not be just cause for delaying testing. At least 75 percent of the time shall be for periods of not less than 4 hours continuous operation.

a. Continuous running at 25 to 50 mph on improved roads for 700 miles at intervals of not less than 4-hour periods

b. Continuous running at 15 to 25 mph on gravel or secondary roads for 150 miles at intervals of not less than 4-hour periods

c. Continuous operation at safe speeds up to 15 mph over off-road terrain for 150 miles at intervals of not less than 4-hour periods

d. Starting the truck with progression through all forward gear ranges 100 times

e. Twenty-five cycles of backing a distance of at least 50 feet at not less than 5 mph, and then applying full brake pressure

f. One-hundred normal stops from a speed of 50 mph

g. Twenty-five quick stops (halting the vehicle within 30 feet from a speed of 20 mph on dry, level concrete or black top surface)

h. Fifty 360° left turns and fifty 360° right turns with 10 left turns and 10 right turns terminating in full brake stops midway through the turn. The turning radius shall not exceed 35 feet. Enter and leave each turn tangentially to the approximate circle generated

i. Ten rocking tests as specified in 4.6.7.2, except for the wide-open throttle shifting portion

j. With the maximum allowable evenly-distributed load secured in the dump body, raise and lower the load from hauling position to maximum dumping angle 125 times

k. Each hydraulic control lever for front-mounted attachments and its corresponding hydraulic system shall be operated through a minimum of 50 cycles. The loading imposed on the hydraulic system during each cycle shall be similar to that imposed during actual plowing operations.

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Note: The items may be conducted concurrently as appropriate.

4.6.8.1 Servicing during test. Only the following types of servicing shall be permitted:

a. Daily servicing (see 6.4) as normally required and performed by the driver

b. Oil change and lubrication at the end of the first 500 miles

c. Such other servicing as recommended and contained in service manuals for this type equipment and for the time or distances specified in such manuals.

4.6.8.1.1 Adjustments. Adjustment of vehicle components shall not be permitted during the endurance tests, except those normally permitted during 500- and 1,000-mile inspections.

4.6.8.1.2 Reconditioning, modification and repair. No reconditioning, modification or repairs shall be accomplished without prior approval of the procuring activity.

4.6.8.2 Inspection after test. After the test, a thorough inspection of the truck shall be accomplished as follows:

a. Major assemblies, such as engine, transmissions, differential, and axles, shall be disassembled as necessary to determine the extent of wear and damage. Erratic operation, overheating, excessive noise, or vibration suggesting potential failures shall be indicative that disassembly is necessary

b. All gears, shafts, bearings, parts, and components disassembled shall be subjected to magnetic particle inspection to determine if cracks have developed through stress

c. Castings used to house major assemblies, or used as part of major assemblies, shall be examined as necessary to determine if damage was incurred during the test.

Cracks or excessive wear damage indicating potential malfunction shall constitute failure of the test.

4.6.9 Electromagnetic interference. The trucks shall be tested for electromagnetic interference in accordance with SAE J551.

4.6.10 Environmental tests. The truck shall be subjected to the following environmental tests in accordance with MIL-STD-810 for storage from -65° to +160°F and operation from -40° to +125°F. After stabilization at operating temperature, the engine shall be started and all operable components shall be operated. The front hitch shall raise and lower (full up and full down) a load equivalent to the maximum loading created by any front mounting attachment specified in 3.5.1 through a minimum of 100 cycles. Each plow attachment control lever shall be operated through a minimum of 100 cycles. The brake system and the steering mechanism shall each be operated through a minimum of 50 cycles. During the steering mechanism portion of the test, the tires shall be rotated a few degrees after each 5 to 10 cycles to prevent excessive tire wear in one spot. When equipped with the dump body, the hydraulic hoist shall raise and lower a maximum allowable load, evenly distributed and secured, from hauling position to

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maximum dumping angle a minimum of 50 times.

4.6.10.1 High temperature. The truck shall be subjected to high temperature storage and operation in accordance with method 501, procedure I. After stabilizing at operating temperature, the unit shall be subjected to tests specified in 4.6.10.

4.6.10.2 Low temperature. The truck shall be subjected to low temperature storage and operation in accordance with method 502, procedure I. The storage temperature shall be maintained for 48 hours. At the conclusion of the storage period, the following operational tests shall be conducted:

4.6.10.2.1 The chamber temperature shall be raised to -40°F and maintained until temperature stabilization of the truck and all its components is reached. The power plant heater(s) shall be connected to 110V power source. The engine(s) shall start within 60 minutes after connecting the heaters. After starting, the truck shall be subjected to the tests specified in 4.6.10. The cab temperature shall be not less than +45°F at the end of 60 minutes of operation of the truck engine and cab heater.

4.6.10.2.2 After the test in 4.6.10.2.1, the chamber temperature shall be raised to -20°F and maintained until temperature stabilization of the truck and all its components is reached. The engine shall start within 5 minutes, using only fluid starting aids and the installed battery. After starting, the truck shall be subjected to the tests specified in 4.6.10. The cab temperature shall be not less than +45°F at the end of the 60 minutes of operation of the truck engine and cab heater.

4.6.10.3 Altitude performance. An evaluation of vehicle characteristics based on actual performance data shall be submitted as part of the test report to show performance attainable at altitude increments of 1,000 feet from sea level to 6,000 feet. For these calculations, use 100 percent of the manufacturer's maximum automotive duty net rating of the engine at sea level and at 60°F.

4.6.11 Maintenance, servicing, and unit replacement test. Maintenance tests shall be conducted to determine the adequacy and ease of performing preventative maintenance, servicing, and unit replacement. The frequency of servicing required for the various components shall be determined by study of existing supplier's recommendations and by servicing required during the preproduction tests. Interferences or obstructions to servicing shall be noted. Any tendency of the vehicle to pick up mud, snow and ice which interferes with servicing and operation shall be noted. Removal and installation of major components and assemblies to demonstrate ease of replacement shall be accomplished. Particular attention shall be given to time required, difficulties encountered, and the tools and skills required in conducting maintenance, servicing, and replacement.

4.6.12 Reliability demonstration and test. Satisfactory completion of all tests specified herein shall be a demonstration of compliance with the quantitative reliability requirements of this specification.

4.6.13 Maintainability evaluation. The truck shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given maintenance with a minimum number of common tools, servicing, and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of dirt, snow, ice, et cetera, that may hinder servicing and operation.

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4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to determine conformance to the requirements of section 5 herein.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with MIL-STD-281, level A, B, or C as specified (see 6.2).

5.2 Unless otherwise specified, the truck shall be shipped as a mobile unit.

5.3 Marking. In addition to any special marking required, marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. When equipped with the proper attachments, the A/S32U-35 truck is intended for runway and general airbase, snow plowing, and ice removal operations. When equipped with the proper attachments, the trucks can also be used for general road maintenance work such as grading and hauling thus providing a year-round vehicle.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When underbody blade is to be furnished (see 3.5.2)

c. When power plant heater kit is not to be installed on the truck engine (see 3.8.2.8)

d. When studded tires are required (see 3.8.6)

e. When emergency air brake system is required (see 3.8.8.3)

f. When dry charged batteries are required (see 3.8.9)

g. When warning lights should be other than yellow (see 3.10.3)

h. Location and conditions for preproduction testing (see 4.4)

i. Applicable levels of preservation, packaging, and packing (see 5.1)

j. When warning lights are other than specified (see 3.10.3a).

6.3 Definitions. For the purpose of this specification, the following definitions will apply:

6.3.1 Reliability. Reliability is defined as the probability of performing a specified function under given conditions without failure for a specified period of time. Recognizing that, in general, the rate of failure of equipment is fairly constant throughout the life of the equipment, the probability of nonfailure over an operating time interval decreases exponentially as a function of the length of the interval, during which time there is a constant failure rate, and can be expressed as follows:

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Confidence level: 90 percent

$$MTBF = \frac{\text{Total test time}}{2.3}$$

$$\text{Reliability} = e^{-x} \text{ where } x = \frac{\text{Mission Time}}{MTBF}$$

The $2.3 = \frac{4.61}{2}$ is based on the constant

for the Poisson/Chi squared distribution, assuming an exponential (2 degrees freedom) distribution even though the failure rate for the test time is zero.

6.3.2 Mean-time-between-failures. The mean life or mean-time-between-failures is the arithmetical mean (average) of the operating time between failures.

6.3.3 Turning radius. The turning radius is defined as the maximum measurement from the center of the turning circle to the centerline of the outside front tire track of the truck.

6.4 Daily servicing. Daily servicing is known in the Armed Services as First Echelon Maintenance (before, during, and after operation check).

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DOCUMENT IDENTIFIER *(Number)* AND TITLE
MIL-T-83118A Truck, Snow Removal Equipment A/S32U-35 (Heavy Duty)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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